Bidirectional Range-Azimuth Upsampling for One-Bit SAR Imaging

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***Abstract*—High-resolution synthetic aperture radar (SAR) produces large raw-data volumes, motivating one-bit acquisition as an extreme form of onboard data reduction. The resulting sign-only echoes, however, lose amplitude information and introduce nonlinear quantization distortion. Existing one-bit SAR methods mainly optimize threshold design or reconstruction, while the allocation of a fixed digital upsampling budget between range and azimuth has received limited attention. This paper proposes bidirectional range-azimuth upsampling (BRAU), which distributes pre-quantization upsampling over both echo dimensions. Experiments on 70 paired samples from seven SAR datasets compare bidirectional and unidirectional allocations under a common separable random threshold. Across five integer sampling budgets, bidirectional allocations consistently provide higher reconstruction fidelity and form a near-optimal region among neighboring factor pairs. Range-compressed spectral measurements also show lower off-support and directional leakage for representative bidirectional processing. These results support two-dimensional spectral redundancy as an operating interpretation of BRAU and provide a practical allocation principle for one-bit SAR imaging under a fixed upsampling budget.**

***Index Terms*—one-bit SAR imaging, range-azimuth allocation, upsampling, quantization**

I. Introduction

Synthetic aperture radar (SAR) forms high-resolution images through coherent processing of echoes collected along a synthetic aperture [1], [2]. Increasing resolution and swath width enlarges the number of complex samples that must be acquired, buffered, processed, and transmitted. This data burden is especially restrictive for spaceborne and airborne platforms, where onboard memory, downlink capacity, and power are limited. Raw-data coding methods such as block-adaptive quantization, flexible dynamic quantization, and predictive coding reduce the bit rate while preserving information required by the imaging processor [3]–[5]. One-bit SAR takes this reduction to the extreme by retaining only the signs of the in-phase and quadrature echo components. Its simple representation can greatly reduce storage and transmission requirements, but it also changes the signal entering every subsequent coherent processing stage.

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Sign quantization discards the continuous amplitude of the complex echo and generates nonlinear harmonics whose spectra can overlap the useful SAR signal [11], [20]. Unlike image-domain compression applied after focusing, one-bit acquisition acts on the two-dimensional raw echo before range compression and azimuth synthesis. The range and slow-time axes therefore determine both the sampled signal support and the placement of quantization distortion. Oversampling can create additional spectral separation before the data are returned to the original imaging grid, but the benefit depends on which dimension is enlarged. Concentrating all additional samples in range changes only the fast-time grid, while azimuth-only processing changes only the slow-time grid. Because SAR focusing uses information from both axes, the dimensional allocation of a limited sampling budget is itself a signal-processing decision.

Early signum-coded SAR studies established that useful images can be formed from binary echoes and investigated real-time processing, interferometry, phase quantization, and oversampling [6]–[11]. In parallel, one-bit compressive-sensing research characterized binary measurements, scale ambiguity, and reconstruction from sign observations [12]–[15]. Model-based and sparsity-aware methods later adapted these ideas to SAR and radar imaging [16]–[18]. This line of work established the feasibility of severe quantization and showed that reconstruction quality depends on both the observation model and the structure imposed during recovery.

Recent one-bit SAR research has placed greater emphasis on the acquisition threshold and suppression of quantization distortion. Time-varying thresholds retain amplitude-related information, single-frequency thresholds relocate harmful harmonics, and low-precision or learning-based methods improve the reconstructed image through refined observation or de-quantization models [19]–[24]. Several of these methods employ oversampling as part of a particular threshold construction, confirming that the sampling grid and threshold spectrum are coupled. Their primary design variable, however, remains the threshold or the reconstruction algorithm. Upsampling is generally prescribed along one dimension, and its range-azimuth allocation is not evaluated under a common resource constraint.

This omission matters when digital upsampling is limited by memory and computation. Two allocations can produce the same number of pre-quantization samples while distributing them differently over the range-azimuth plane. Treating them as equivalent ignores the two-dimensional spectral support of SAR echoes and the directional structure of quantization distortion. It also leaves open a practical question: for a fixed increase in sample count, should the entire budget be assigned to one axis or shared by both? Answering this question requires controlled comparisons with the same threshold, data, image-formation chain, region of interest, and normalization protocol.

Let R and A denote the range and azimuth upsampling factors, respectively, and let $Q_{\mathrm{up}} = RA$ be the total pre-quantization allocation. Under the same Q_{up}, a unidirectional strategy assigns the full budget to one dimension, whereas a bidirectional strategy uses $R > 1$ and $A > 1$. This paper formulates bidirectional range-azimuth upsampling (BRAU) and evaluates whether dimensional allocation affects one-bit SAR reconstruction. BRAU operates on the complex raw echo before quantization and is distinct from interpolation of an already reconstructed image. The main contributions are as follows:

- We formulate fixed-budget range-azimuth upsampling allocation before one-bit quantization, distinguishing bidirectional raw-echo processing from post-reconstruction image interpolation.
- We show on 70 paired samples that bidirectional allocations consistently outperform unidirectional allocations under a common separable random threshold, with a near-optimal plateau among several bidirectional factor pairs.
- We provide range-compressed spectral evidence showing that bidirectional allocation reduces off-support and directional leakage relative to unidirectional allocation.

II. Method

A. Fixed-Budget Range-Azimuth Upsampling

Let $s(\tau, \eta) \in \mathbb{C}$ denote the complex raw SAR echo, where τ and η are range time and azimuth time. Writing $s = I + jQ$, one-bit quantization is applied separately to the in-phase and quadrature components:

$$y = \mathrm{sign}(\mathrm{Re}(s+u)) + j\,\mathrm{sign}(\mathrm{Im}(s+u)), \qquad (1)$$

where $u \in \mathbb{C}$ is the quantization threshold. The sign operation removes continuous amplitude information and introduces nonlinear harmonic distortion.

BRAU allocates a fixed digital upsampling budget

$$Q_{\mathrm{up}} = R \times A \qquad (2)$$

between range and azimuth. Range-only and azimuth-only allocations are $(Q_{\mathrm{up}}, 1)$ and $(1, Q_{\mathrm{up}})$, respectively, while a bidirectional allocation satisfies $R > 1$ and $A > 1$. The $(1,1)$ case is the no-upsampling baseline.

For each allocation, the raw echo is transformed along the selected dimension, centered with `fftshift`, zero-padded symmetrically, and transformed back to the enlarged grid. The threshold is generated on this grid, followed by one-bit quantization and range compression. When range is enlarged, the range sampling frequency and range-time grid are updated consistently. After range compression, frequency-domain cropping returns the data to the original grid, after which all methods use identical range cell migration correction (RCMC), image formation, region of interest (ROI), and normalization. Following the frequency-domain interpolation procedures adopted in previous one-bit SAR studies [20], [24], this implementation uses centered FFT zero-padding rather than regenerating the echo through a new SAR simulation. BRAU is therefore a pre-quantization raw-data operation, not image-domain interpolation.

TABLE I
BRAU PROCESSING PIPELINE.

Step	Operation
1	Select (R, A) under $Q_{\mathrm{up}} = RA$.
2	Apply centered FFT zero-padding along range and azimuth.
3	Generate the threshold on the enlarged echo grid.
4	Quantize the in-phase and quadrature components to one bit.
5	Perform range compression with the matched sampling parameters.
6	Crop the range-compressed data to the original grid in frequency.
7	Apply common RCMC, image formation, ROI, and normalization.

B. Threshold Setting

The main allocation experiment uses a separable random threshold (RT), following the fluctuating-threshold design principle used in one-bit SAR imaging [24]. Its phase is the sum of independent range and azimuth phase vectors:

$$u_{\mathrm{RT}}(m, n) = A_s \hat{\sigma} \exp\left[j\left(\phi_r(m) + \phi_a(n)\right)\right], \qquad (3)$$

where $\hat{\sigma} = \sqrt{2/\pi}\,\mathrm{mean}(|s_{\uparrow}|)$ is the scale estimated from the upsampled echo, and A_s controls the threshold amplitude. The fixed-budget results use $A_s = 0.6$ as a common comparison setting.

C. Evaluation Protocol

The experiments use seven SAR datasets representing different land-cover and structural patterns. Ten stratified windows are selected from each dataset, giving 70 paired samples. The ground truth (GT) is reconstructed from the unquantized complex echo using the same range compression, RCMC, image formation, ROI, and normalization as the one-bit results. RT is evaluated with seed 2026 and the common amplitude $A_s = 0.6$ for $Q_{\mathrm{up}} \in \{4, 6, 8, 9, 10\}$.

Reconstruction fidelity is evaluated using peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) [25]. Equivalent number of looks (ENL) is evaluated on intensity, obtained by squaring the normalized amplitude. For a homogeneous intensity region, ENL is defined as [26]

$$\mathrm{ENL} = \frac{\mu_I^2}{\sigma_I^2}, \qquad (4)$$

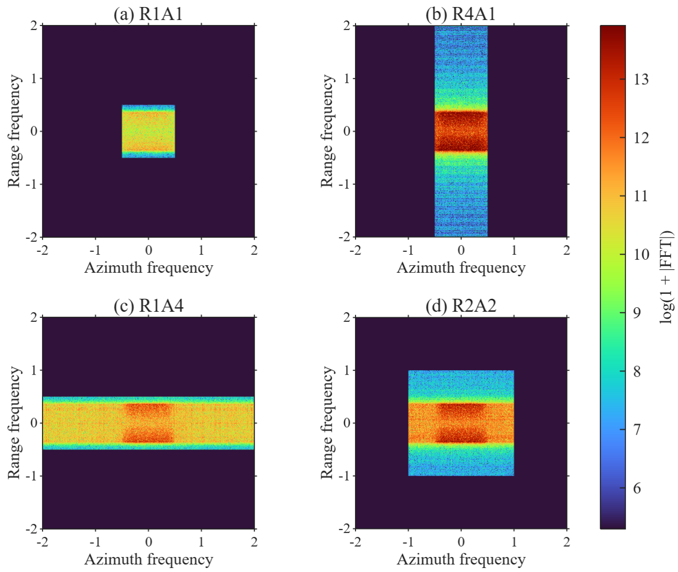

Fig. 1. Range-compressed spectra for representative $Q_{\mathrm{up}} = 4$ allocations on a common 4×4 normalized-frequency canvas. All panels use the same color scale. R4A1 and R1A4 extend one spectral dimension, whereas R2A2 distributes the extension over both dimensions.

where μ_I and σ_I^2 are the mean and variance within a homogeneous ROI. ENL is calculated for 20 samples from the field and port datasets. A 64×64 ROI is selected on each GT image and then reused unchanged for every allocation. PSNR and SSIM use all 70 samples.

For each budget, the best bidirectional and unidirectional groups are selected by mean PSNR. Paired Wilcoxon signed-rank tests [27] are applied to the corresponding 70-sample vectors.

III. Experiments and Results

A. Mechanism Analysis Through Spectral Leakage

One-bit quantization is known to generate nonlinear harmonic distortion [11], [20], [23]. In the two-dimensional SAR data considered here, this distortion appears along both range and azimuth. The working interpretation of BRAU is that the effective SAR echo occupies a two-dimensional range-azimuth spectral support, whereas a unidirectional allocation enlarges only one spectral axis. In contrast, a bidirectional allocation supplies frequency-domain redundancy in both dimensions and can reduce the overlap between quantization distortion and the effective support. Fig. 1 compares representative allocations for $Q_{\mathrm{up}} = 4$ on the common frequency range $[-2, 2] \times [-2, 2]$.

For an intermediate matrix X, define the spectral energy

$$S_X(k_r, k_a) = |\mathcal{F}_2\{X\}(k_r, k_a)|^2. \tag{5}$$

The support mask M is obtained from a reference spectrum using a relative magnitude threshold $\tau = 0.35$. The off-support energy ratio is

$$\rho_{\mathrm{off}} = \frac{\sum S_X(1 - M)}{\sum S_X}. \tag{6}$$

TABLE II
Spectral Leakage After Range Compression for $Q_{\mathrm{up}} = 4$. Lower is Better.

Allocation	Off-support	Range leakage	Azimuth leakage
R1A1	0.926008	0.471044	0.101785
R4A1	0.893954	0.368013	0.070476
R1A4	0.901018	0.394427	0.072251
R2A2	**0.892857**	**0.363750**	**0.066680**

Directional leakage is computed from the projected spectra

$$P_r(k_r) = \sum_{k_a} S_X(k_r, k_a), \quad P_a(k_a) = \sum_{k_r} S_X(k_r, k_a), \tag{7}$$

and the projected masks

$$M_r(k_r) = \max_{k_a} M(k_r, k_a), \quad M_a(k_a) = \max_{k_r} M(k_r, k_a). \tag{8}$$

The resulting ratios are

$$\rho_r = \frac{\sum P_r(1 - M_r)}{\sum P_r}, \qquad \rho_a = \frac{\sum P_a(1 - M_a)}{\sum P_a}. \tag{9}$$

As shown in Table II, all upsampled groups reduce leakage relative to R1A1, and R2A2 obtains the lowest off-support, range-direction, and azimuth-direction ratios. This representative result supports the interpretation that bidirectional allocation provides two-dimensional spectral redundancy and reduces the overlap between one-bit quantization distortion and the effective SAR support. The metrics are reported at the range-compression stage; no additional improvement is attributed to RCMC because it does not materially change these spectral-energy ratios.

B. Threshold Design

Prior one-bit SAR studies show that threshold amplitude influences reconstruction quality [19], [24]. Fig. 2 shows the RT amplitude sensitivity for the best bidirectional allocations at $Q_{\mathrm{up}} = 4, 6, 9$. The SSIM first increases and then decreases with A_s. The best tested values shift from $A_s = 0.7$ for $Q_{\mathrm{up}} = 4$ to 0.8 for $Q_{\mathrm{up}} = 6$ and 1.0 for $Q_{\mathrm{up}} = 9$. The common $A_s = 0.6$ used in the main experiment is therefore a controlled setting for allocation comparison rather than a global optimum for every budget.

C. Main Fixed-Budget Results

Fig. 3 shows a representative scene using a common normalized-amplitude scale. Both unidirectional allocations improve over one-bit imaging without upsampling, while R2A2 more closely follows the GT in extended boundaries and local scattering texture. The shared $[0, 1]$ color limit makes the four panels directly comparable.

Table III reports the RT results under the common $A_s = 0.6$ setting. R1A1 gives the lowest reconstruction fidelity, whereas a bidirectional group reaches the highest PSNR and SSIM at every listed budget. At $Q_{\mathrm{up}} = 4$, R2A2 obtains 26.0623 dB and 0.8026. At $Q_{\mathrm{up}} = 6$, R2A3 gives the highest PSNR of 26.5585 dB, while R3A2 gives the highest SSIM of 0.8246. At $Q_{\mathrm{up}} = 8$, R2A4 reaches 26.8194 dB and 0.8363. R3A3 is

TABLE III
FIXED-BUDGET EVALUATION METRICS UNDER SEPARABLE RT WITH $A_s = 0.6$.

Allocation	PSNR	SSIM	ENL
GT	∞	1.0000	1.0321
R1A1	23.0517	0.6359	1.0094
$Q_{\mathrm{up}} = 4$			
R4A1	25.7767	0.7857	1.0202
R1A4	25.7258	0.7843	1.0174
R2A2	**26.0623**	**0.8026**	**1.0217**
$Q_{\mathrm{up}} = 6$			
R6A1	26.1032	0.8038	1.0161
R1A6	26.1415	0.8037	1.0173
R2A3	**26.5585**	0.8246	1.0175
R3A2	26.5525	**0.8246**	**1.0204**
$Q_{\mathrm{up}} = 8$			
R8A1	26.3618	0.8148	1.0173
R1A8	26.3358	0.8136	1.0179
R2A4	**26.8194**	**0.8363**	1.0183
R4A2	26.8066	0.8356	**1.0187**
$Q_{\mathrm{up}} = 9$			
R1A9	26.3978	0.8171	1.0174
R3A3	**26.9167**	**0.8400**	**1.0181**
R9A1	26.4482	0.8186	1.0180
$Q_{\mathrm{up}} = 10$			
R1A10	26.4481	0.8198	**1.0192**
R2A5	26.9418	0.8423	1.0171
R5A2	**26.9702**	**0.8427**	1.0173
R10A1	26.4630	0.8206	1.0171

PSNR and SSIM are averaged over 70 paired samples; ENL uses 20 fixed 64×64 homogeneous ROIs.

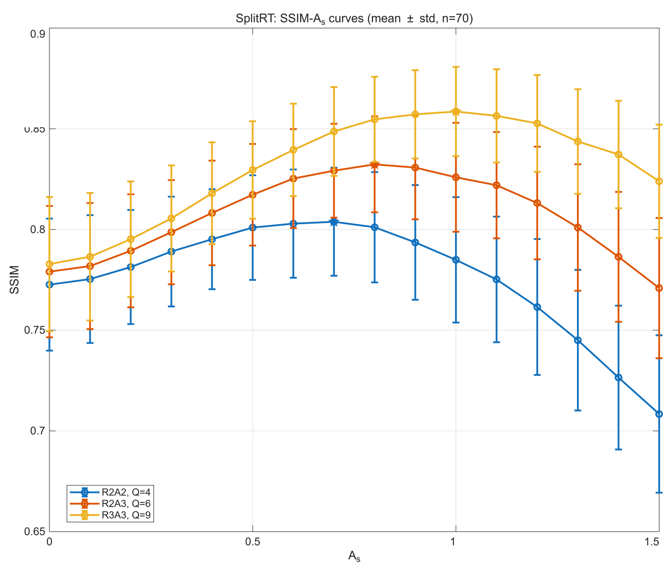

Fig. 2. Mean SSIM of RT versus A_s for representative bidirectional allocations.

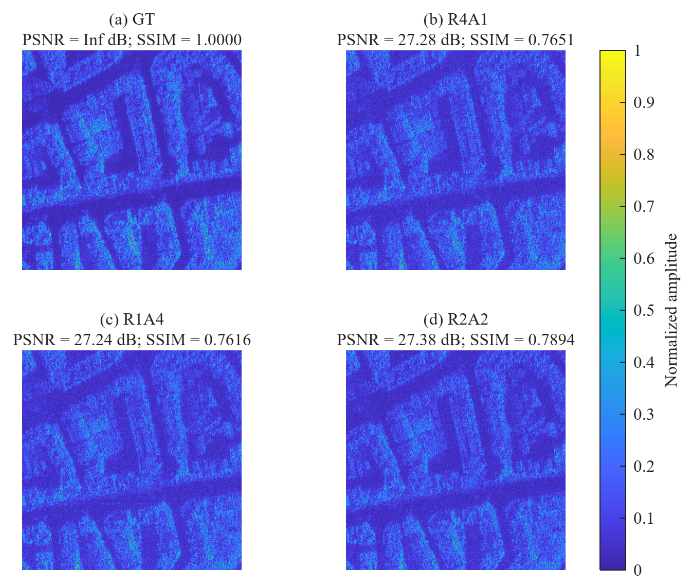

Fig. 3. Representative normalized-amplitude SAR images for $Q_{\mathrm{up}} = 4$ with sample-specific PSNR and SSIM relative to GT. GT is self-compared; all panels share the same $[0, 1]$ color scale.

best at $Q_{\mathrm{up}} = 9$ with 26.9167 dB and 0.8400, and R5A2 is best at $Q_{\mathrm{up}} = 10$ with 26.9702 dB and 0.8427.

Across the five integer budgets, the mean-best bidirectional allocations improve PSNR by 0.2856–0.5072 dB and SSIM by 0.0169–0.0221 over the mean-best unidirectional allocations. Paired comparisons on the corresponding 70-sample vectors give $p_{\mathrm{PSNR}} \leq 4.34 \times 10^{-10}$ and $p_{\mathrm{SSIM}} \leq 3.56 \times 10^{-13}$. The close results of R2A3 and R3A2 at $Q_{\mathrm{up}} = 6$, R2A4 and R4A2 at $Q_{\mathrm{up}} = 8$, and R2A5 and R5A2 at $Q_{\mathrm{up}} = 10$ further indicate a near-optimal region rather than a unique factor pair.

ENL varies less across allocations and is reported as a descriptive speckle statistic rather than a primary ranking criterion. Its maximum coincides with the fidelity-optimal group at $Q_{\mathrm{up}} = 4$ and 9, while R1A10 has the largest ENL at $Q_{\mathrm{up}} = 10$. Together, the PSNR and SSIM results indicate that bidirectionality, rather than exact balance, is the main allocation property under the separable RT.

IV. CONCLUSION

This paper studied how a fixed pre-quantization digital upsampling budget should be divided between range and azimuth in one-bit SAR imaging. Under a common separable random threshold, bidirectional allocations consistently achieved higher reconstruction fidelity than concentrating the same budget in one dimension, while neighboring bidirectional factor pairs formed a near-optimal plateau. Range-compressed spectral measurements further showed that bidirectional processing reduced off-support and directional leakage. BRAU therefore provides a fixed-budget allocation principle that exploits the two-dimensional spectral redundancy of SAR echoes before one-bit quantization.

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